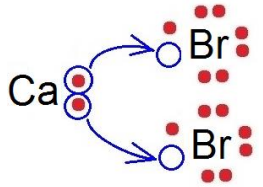


Section A

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
1.				 (1) forming Ca^{2+} and Br^- ions (1)						
2.				sodium and oxygen (1) they have the largest electronegativity difference (1)	1	1		2		
3.	(a)			in a coordinate bond both bonding electrons come from one atom while in a covalent bond both atoms contribute one electron in the shared pair	1			1		
	(b)			e.g. Al_2Cl_6 , NH_4^+	1			1		
4.	(a)			${}^{81}_{36}\text{Kr} + {}^0_{-1}\text{e} \rightarrow {}^{81}_{35}\text{Br} (+\text{v})$		1		1		
	(b)			2.29×10^5 years		1		1	1	
5.				$10.8 = (11 \times 0.8) + (x \times 0.2)$ (1) $0.2x = 2$ $x = 10$ (1)		2		2	1	
Section A total					3	7	0	10	2	0